

ChE 451 (Renewable Energy Technologies)
Department of Chemical Engineering
Fall 2025

Instructor:

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Office Hours: By appointment

Classroom (in person): EIB 271 (945 W Taylor Street, Chicago, IL 60607)

Prerequisite: course suitable for senior and graduate engineering, science, and business students.

Course outline

Part I: Sources of Renewable Energy

- 1) Solar
- 2) Wind
- 3) Biomass
- 4) Other sources

Part II: Energy Storage and Conversion

- 5) Batteries and Fuel Cells

Part III: The Hydrogen Economy

- 6) Hydrogen Production (reforming and electrolysis)
- 7) Hydrogen Storage and Transportation

Part IV: Case Studies (Applications)

- 8) Hybrid Wind/Solar Hydrogen Systems
- 9) Hybrid Desalination Systems: fuel cell/desalination, solar desalination

Text Books:

“Hybrid Hydrogen Systems for Stationary and Transportation Applications”, Said Al-Hallaj and Kristofer Kiszynski, 1st Edition, 2011, Springer Publishing, ISBN 978-1-84628-466-3

Grade:

- 35% Weekly assignments (3-4 mini-projects: written essays and engineering design calculations)
- 25% Midterm Home Exam
- 40% Final Project: written report (20%) and oral presentation (20%)
Project Grading Criteria: originality, approach, content and relation to course material

Class Schedule

<u>Week</u>	<u>Topic</u>
8/25	Introduction (Renewable Energy and Sustainability)
9/1	Labor Day Holiday
9/8	Solar Energy Calculations and Solar Photovoltaics (PV) (<i><u>Discussion of Final Project</u></i>)
9/15	Wind Energy Basics and Applications
9/22	Bioenergy (<i><u>Final Project Abstract Due</u></i>)
9/29	Energy Storage for Smart Grid Applications
10/6	Safety and Thermal Management of Lithium-ion Batteries
10/13	Thermal Energy Storage
10/20	Solar + Storage Project Development (<i>Guest Lecture: Jake Edie, Clean Energy Associates</i>)
10/27	Midterm (<i><u>Final Project Pitch</u></i>)
11/3	Hydrogen Economy and Hydrogen Storage
11/10	Hydrogen applications (Gues Speaker: TBD)
11/17	Fuel Cells
11/24	Case Studies: Water Desalination
12/1	Final (oral presentation)
12/8	Finals Week (written report due)